(14)

Kaushik R
EE20B057

Solid State Devices (EE3001)
Quiz#1 (Total Mark 15)

Date: 23rd Feb., 2022 **Time: 8 am to 8:50 am**

Answer all questions. Final answers must be written in the space provided after each question. Marks will be deducted if proper units are not mentioned. All important steps to arrive at the final answer must be shown in the additional answer sheet.

Assume intrinsic carrier concentration (n_i) in Si is 10^{10} /cm^3 and kT = 26 meV at T = 300K. Bandgap of Si can be assumed as 1.1 eV.

Q1. A semiconductor-X with E_g = 1.5 eV has the same values of N_C and N_V as silicon.

(a) The value of n_i at 300 K for this semiconductor-X is 4.5632 x 10^6 /cc (2) [2]

(b) The value of n_i at 500 K for this semiconductor-X is 8.8077 x 10^{11} /cc X (1) [2]

Q2. A particular semiconductor sample is doped with 2×10^{15} donor atoms/cm^3. The intrinsic carrier concentration (n_i) in the semiconductor is 10^{13} /cm^3 at 300 K. Assume that n_i changes by one order of magnitude for every 100 K and complete ionization of the impurities.

(a) Position of the Fermi level at 300 K is $E_F = E_I +$ 0.138 eV (1) [1]

(b) Position of the Fermi level at 500 K is $E_F = E_I +$ 0.0382 eV (2) [2]

Q3. The donor impurity concentration in a Si sample (sample-A) is given by N_D = 4x10^{16} /cm^3. Now additional acceptor doping of N_A = 6x10^{16} /cm^3 is introduced (sample-B). Assume complete ionization and that due to the doping variation, electron and hole mobilities are not changed and electron mobility is three times of the hole mobility throughout.
Ratio of the resistivity of the sample-A (ρ_A) to that of sample-B (ρ_B) is ρ_A/ρ_B = 1/6 [2] (2)

Q4. The position of the Fermi level for a silicon sample is at $E_f = E_i$ + 0.45 eV at 300 K. The sample is uniformly illuminated, thereby generating an additional 10^{19} electron-hole pairs per cm^3 per second. Assume n_i for Si is 10^{10} /cm^3 and kT=26 meV at 300 K. If the positions of the hole quasi Fermi levels at 300 K for this illuminated sample appears to be 0.2 eV below E_i, the minority carrier lifetime in this sample is 2.191 x 10^{-6} s (3) [3]

Q5. Light is shone on the surface of an n-type silicon wafer doped with 10^{17} donor atoms/cm^3 generating electron-hole pairs only on the surface. Assume T = 300K, kT/q = 26 mV, μ_n = 1350 cm^2/V-sec, μ_p = 450 cm^2/V-sec, τ_n = τ_p = 1 μs. If the hole concentration (p_n) is 10^{14}/ cm^3 at 1 μm from the surface, the value of p_n at a distance of 20 μm from this surface is 5.14 x 10^{13} /cc (3) [3]

Q1. A semiconductor-X with E_g = 1.5 eV has the same values of N_C and N_V as silicon.

(a) The value of n_i at 300 K for this semiconductor-X is 4.5632×10^6 /cc ✓ (2) [2]

(b) The value of n_i at 500 K for this semiconductor-X is 8.8077×10^{11} /cc ✗ (1) [2]

a) $n_i = \sqrt{N_C N_V}\exp\left(\frac{-E_g}{2KT}\right)$ 1.1

$= \sqrt{N_C N_V}\exp\left(\frac{-1.1}{2KT}\right).\exp\left(\frac{-0.2}{0.026}\right)$

$= 4.56 \times 10^6/cm^3$

b) $\frac{n_i(500)}{n_i(300)} = \left(\frac{500}{300}\right)^{3/2}\exp\left(\frac{-E_g}{2K}\left(\frac{1}{500}-\frac{1}{300}\right)\right)$

$n_i(500) = 4.56\times10^6 \times \left(\frac{5}{3}\right)^{1.5}\exp\left(\frac{-1.5}{0.052}\left(\frac{-2}{5}\right)\right) = 1.006\times10^{12}/cm^3$

Q2. A particular semiconductor sample is doped with 2×10^{15} donor atoms/cm^3. The intrinsic carrier concentration (n_i) in the semiconductor is 10^{13} /cm^3 at 300 K. Assume that n_i changes by one order of magnitude for every 100 K and complete ionization of the impurities.

(a) Position of the Fermi level at 300 K is $E_F = E_I +$ 0.1378 eV ✓ (1) [1]

(b) Position of the Fermi level at 500 K is $E_F = E_I +$ 0.0382 eV ✓ (2) [2]

a) $N_D = 2\times10^{15}$ $n_i = 10^{13}$

$n = 2\times10^{15} = 10^{13}\exp\left(\frac{E_F - E_i}{0.026}\right) \Rightarrow E_F - E_i = 0.1378$

b) $n = p + 10^{15}\times2$ $np = 10^{30}$ $n^2 - 2\times10^{15}n - 10^{30} = 0$ $n = 10^{15}(1+\sqrt{2})$

$10^{15}\times2.414 = 10^{15}\exp\left(\frac{E_F - E_i}{0.026\times5/3}\right)$ $p = 10^{15}(\sqrt{2}-1)$

$E_F - E_i = 0.0382$ eV

Q3. The donor impurity concentration in a Si sample (sample-A) is given by $N_D = 4\times10^{16}$ /cm^3. Now additional acceptor doping of $N_A = 6\times10^{16}$ /cm^3 is introduced (sample-B). Assume complete ionization and that due to the doping variation, electron and hole mobilities are not changed and electron mobility is three times of the hole mobility throughout.

Ratio of the resistivity of the sample-A (ρ_A) to that of sample-B (ρ_B) is $\rho_A/\rho_B =$ 1/6 ✓ (2) [2]

$n_A = 4\times10^{16}$ $\frac{\rho_A}{\rho_B} = \frac{\sigma_B}{\sigma_A} = \frac{p_B \mu_p}{n_A \mu_n} = \frac{1}{3}\times\frac{1}{2} = \frac{1}{6}$

$n + 6\times10^{16} = p + 4\times10^{16} \Rightarrow p = 2\times10^{16}$

Q4. The position of the Fermi level for a silicon sample is at $E_f = E_i + 0.45$ eV at 300 K. The sample is uniformly illuminated, thereby generating an additional 10^{19} electron-hole pairs per cm^3 per second. Assume n_i for Si is 10^{10} /cm^3 and kT=26 meV at 300 K. If the positions of the hole quasi Fermi levels at 300 K for this illuminated sample appears to be 0.2 eV below E_i, the minority carrier lifetime in this sample is 2.191 x 10^{-6} / s (3) [3]

$$n = 10^{10} \exp\left(\frac{0.45}{0.026}\right) = 3.285 \times 10^{17}$$

$$G = 10^{19} \qquad p = 10^{19}\tau$$

$$E_i - E_{fp} = 0.2 \qquad 10^{19}\tau = 10^{10} \exp\left(\frac{0.2}{0.026}\right)$$

$$\tau = 2.19 \times 10^{-6}\ s$$